

JSS2 Basic Science Diagnostic Assessment (Part 3)

Curriculum: Cambridge Lower Secondary (Stages 7 & 8) & Nigerian JSS2
Cluster 3: Physics and Scientific Inquiry (Questions 41 - 60)

Science - Cluster 3: Physics & Inquiry

No.	Question & Options	Ans	Diagnostic Focus / Concept Tag
41	What is the standard unit of Force? A) Joule B) Watt C) Newton D) Pascal	C	Fundamental units (Physics).
42	Which type of force opposes the motion of one surface over another? A) Gravity B) Magnetism C) Friction D) Electrostatic	C	Interaction of forces.
43	A car travels 150 km in 3 hours. What is its average speed? A) 50 km/h B) 450 km/h C) 0.02 km/h D) 153 km/h	A	Speed, Distance, Time calculation.
44	Energy can neither be created nor destroyed; it only changes form. This is known as the Law of: A) Gravity B) Conservation of Energy C) Relativity D) Motion	B	Foundational Physics Laws.
45	Which of these is a renewable energy source? A) Coal B) Natural Gas C) Solar D) Crude Oil	C	Energy resources.
46	Identify the method of heat transfer through a solid metal rod: A) Convection B) Radiation C) Conduction D) Evaporation	C	Thermal Physics (Heat transfer).
47	Which component in a circuit provides the electrical energy? A) Switch B) Wire C) Cell (Battery) D) Bulb	C	Basic Electricity.
48	Like magnetic poles (e.g., North and North) will: A) Attract B) Neutralize C) Repel D) Melt	C	Magnetism properties.
49	Pressure is defined as Force divided by: A) Mass B) Velocity C) Area D) Volume	C	Pressure concepts (JSS2 Mechanics).
50	When light passes from air into water, it bends. This is called: A) Reflection B) Absorption C) Refraction D) Dispersion	C	Optics (Light behavior).
51	Which instrument is used to measure current in a circuit? A) Voltmeter B) Ammeter C) Barometer D) Thermometer	B	Electrical measurements.
52	A lever is an example of a: A) Complex engine B) Simple machine C) Force vector D) Energy sink	B	Work and Machines.
53	Which planet is known as the "Red Planet"? A) Venus B) Jupiter C) Mars D) Saturn	C	Space Science (Astronomy).
54	In a scientific experiment, the factor that is changed by the investigator is the: A) Control variable B) Independent variable C) Dependent variable D) Constant	B	Scientific Method / Inquiry.
55	A possible explanation for an observation that can be tested is a: A) Conclusion B) Law C) Hypothesis D) Result	C	Scientific Inquiry skills.
56	Which of these materials is the best thermal insulator? A) Copper B) Aluminum C) Wool (Plastic/Air) D) Iron	C	Properties of materials.

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57	Echoes are caused by the _____ of sound waves. A) Refraction B) Reflection C) Creation D) Absorption	B	Sound Physics.
58	The tilting of the Earth on its axis causes: A) Tides B) Days and Nights C) Seasons D) Moon phases	C	Earth Science/Space.
59	Which safety device breaks a circuit if the current is too high? A) Resistor B) Transformer C) Fuse D) Generator	C	Practical Electricity/Safety.
60	Diagnostic Insight: Difficulty with Q43 and Q49 suggests weakness in applying Math to Science (Quantitative reasoning), essential for SS1 Physics.	-	Pathway alignment.